

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	A.Bhavana	Reg. No.:	192472141
Section / Batch:	CSE-AI	Date:	

Code of Conduct

I A. Bhavana (192472141) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A. Bhavana

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Annexure A – Application Based Questions

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.

Identify email addresses.

Extract mobile numbers.

Detect technical skills (Python, Java, SQL, Machine Learning, NLP).

Extract years of experience.

Generate a structured summary of the candidate's profile.

Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

```
import re
resume = """
Name: Gayathri P
Email: gayathri@gmail.com
Phone: 9876543210
Skills: Python, Java, SQL, Machine Learning, NLP
Experience: 3 years of experience as a Python Developer.
name = re.search(r"Name\s*:\s*(.*)", resume)
if name:
    name = name.group(1)
else:
    name = "Not Found"
email = re.search(r"[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}", resume)
if email:
    email = email.group()
else:
    email = "Not Found"
phone = re.search(r"\b\d{10}\b", resume)
if phone:
    phone = phone.group()
else:
    phone = "Not Found"
skill_list = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
skills = []
for skill in skill_list:
    if re.search(skill, resume, re.IGNORECASE):
        skills.append(skill)
exp = re.search(r"(\d+)\s+years?", resume, re.IGNORECASE)
if exp:
    experience = int(exp.group(1))
else:
    experience = 0
print("----- Candidate Profile -----")
print("Name      :", name)
print("Email     :", email)
print("Phone     :", phone)
print("Skills    :", ", ".join(skills))
print("Experience :", experience, "Years")
print("\n----- Eligibility Status -----")
if experience >= 2 and "Python" in skills:
    print("Candidate is ELIGIBLE for shortlisting.")
else:
    print("Candidate is NOT ELIGIBLE.")
```

OUTPUT:

```
*** ----- Candidate Profile -----
Name       : A Bhavana
Email      : bhavana@gmail.com
Phone      : 9866415444
Skills     : Python, Java, SQL, Machine Learning, NLP
Experience : 3 Years

----- Eligibility Status -----
Candidate is ELIGIBLE for shortlisting.
```

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.
Perform prefix-based searches.
Perform suffix-based searches.
Search using partial keywords.
Perform case-insensitive searches.
Display all matching product names.
Generate a report showing the total number of matching products for each search.

```
import re

products = [
    "Laptop",
    "Laptop Bag",
    "Mouse",
    "Wireless Mouse",
    "Keyboard",
    "Gaming Keyboard",
    "Monitor",
    "Mobile Phone",
    "Phone Charger",
    "Smart Watch"
]

keyword = input("Enter search keyword: ")

# Exact Search
exact = [p for p in products if re.fullmatch(keyword, p, re.IGNORECASE)]

# Prefix Search
prefix = [p for p in products if re.match(keyword, p, re.IGNORECASE)]

# Suffix Search
suffix = [p for p in products if re.search(keyword + r"$", p, re.IGNORECASE)]

# Partial Search
partial = [p for p in products if re.search(keyword, p, re.IGNORECASE)]

print("\nExact Match:")
print(exact)

print("\nPrefix Match:")
```

```

print(prefix)

print("\nSuffix Match:")
print(suffix)

print("\nPartial Match:")
print(partial)

print("\n----- Report -----")
print("Exact Matches:", len(exact))
print("Prefix Matches:", len(prefix))
print("Suffix Matches:", len(suffix))
print("Partial Matches:", len(partial))
OUTPUT:

```

```

.. Enter product keyword: book

----- Product Search Report -----

Exact Match:
No Match

Prefix Match:
No Match

Suffix Match:
Python Programming Book
Java Programming Book

Partial Match (Case-Insensitive):
Python Programming Book
Java Programming Book

----- Search Summary -----
Exact Matches      : 0
Prefix Matches     : 0
Suffix Matches     : 2
Partial Matches    : 2

```

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

- Validate the student register number.
- Validate the institutional email address.
- Validate the course code.
- Validate the semester information.
- Validate the student's mobile number.
- Display appropriate validation messages for each field.
- Generate a final registration status report indicating whether the registration is successful.

```

import re

reg_no = input("Enter Register Number: ")
email = input("Enter Institution Email: ")
course = input("Enter Course Code: ")
semester = input("Enter Semester: ")
mobile = input("Enter Mobile Number: ")

status = True

# Register Number (Example: 22CS101)
if re.fullmatch(r"\d{2}[A-Z]{2}\d{3}", reg_no):
    print("Register Number: Valid")
else:
    print("Register Number: Invalid")
    status = False

```

```

# Institution Email
if re.fullmatch(r"[a-zA-Z0-9._%+-]+@college\.edu", email):
    print("Email: Valid")
else:
    print("Email: Invalid")
    status = False

# Course Code (Example: CS501)
if re.fullmatch(r"[A-Z]{2}\d{3}", course):
    print("Course Code: Valid")
else:
    print("Course Code: Invalid")
    status = False

# Semester (1-8)
if re.fullmatch(r"[1-8]", semester):
    print("Semester: Valid")
else:
    print("Semester: Invalid")
    status = False

# Mobile Number
if re.fullmatch(r"[6-9]\d{9}", mobile):
    print("Mobile Number: Valid")
else:
    print("Mobile Number: Invalid")
    status = False

print("\n----- Registration Status -----")
if status:
    print("Registration Successful")
else:
    print("Registration Failed")

```

OUTPUT:

```

*** Enter Register Number: 192472141
Enter Institution Email: 192472141.simats@saveetha.com
Enter Course Code: DSA0305
Enter Semester: 3RD
Enter Mobile Number: 9866415444
Register Number: Invalid
Email: Invalid
Course Code: Invalid
Semester: Invalid
Mobile Number: Valid

----- Registration Status -----
Registration Failed

```

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____